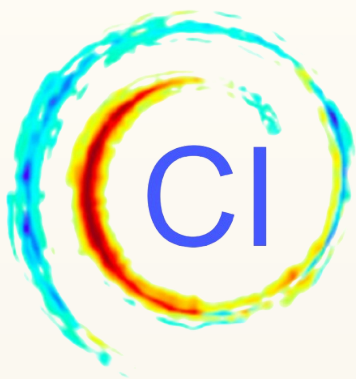




Climate Informatics 2026
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Topography-aware Temperature Forecasting in Europe

Frameworks for introducing topography priors into forecasting models

Laboratory of Environmental Computational Science and Earth Observation (ECEO), EPFL

In Short: Forecasting models such as ViT, trained on satellite data of the predicted modality, are **sensitive to topographical bias**. This work proposes three different frameworks to induce a topography prior and mitigate the observed bias. We show that all three result in a bias reduction, noting that the biggest gap yields from a good **neural representation of the geography**. Furthermore, we provide a **Regularization Framework Template (EECR)** that can be extended for different applications, and provide examples of instantiation where the regularization further improves prediction accuracy and alpine bias mitigation.



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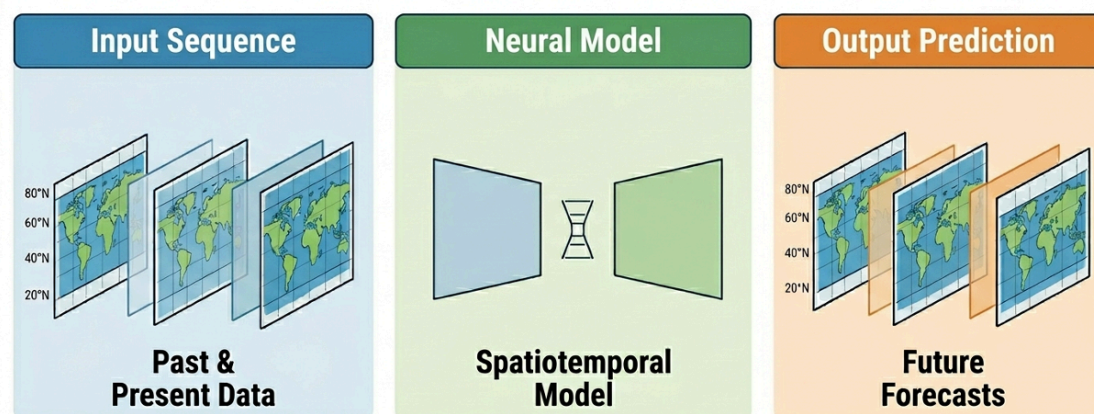
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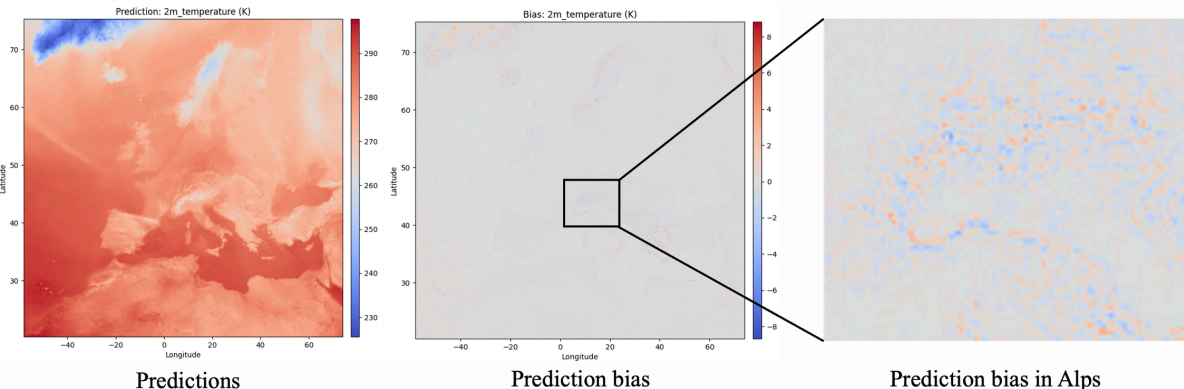
Introduction

Despite recent advances in deep learning for weather forecasting, current models still suffer from significant topography related bias

To address this issue, we investigate three different approaches to **inform the weather forecasting model of topography prior**



Weather Forecasting



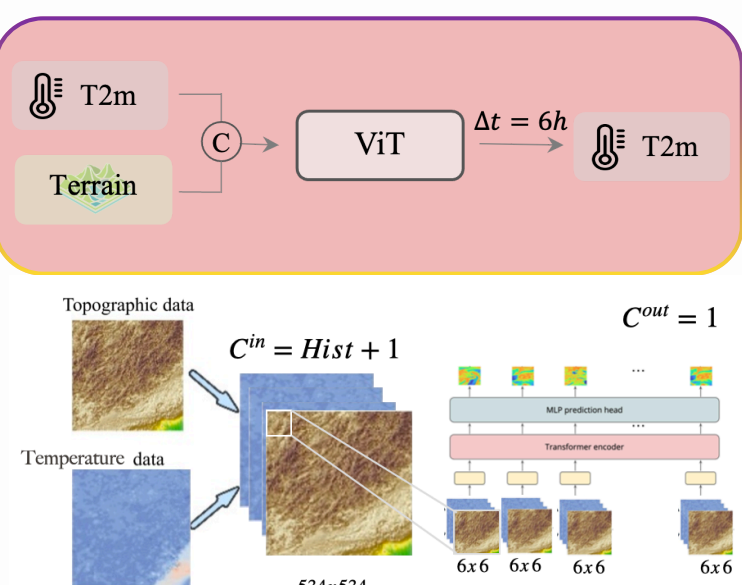
Bias in Alpine Region

Simple Fusion

Motivation
Topography can be injected as additional input variables

How
Topography information (DEM) is concatenated together with meteorological variables and processed by the forecasting model at the input

Limitation
Lack of explicit cross-modal interactions between DEM and meteorological variables

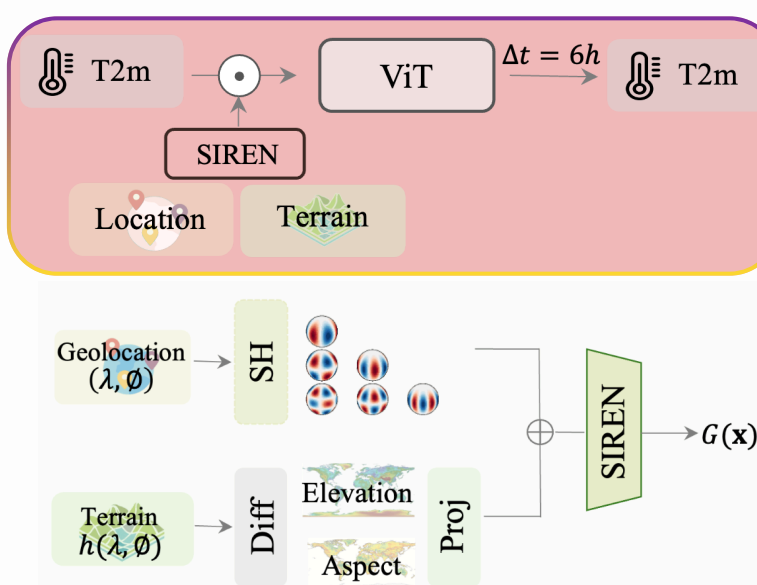


Geo Implicit Neural Representation

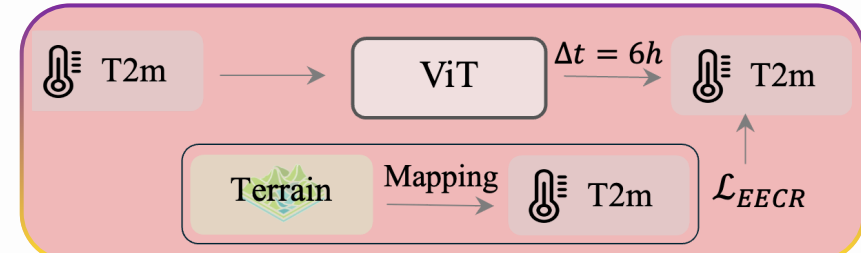
Motivation
Topography can be projected as positional embeddings to inform the forecasting in the latent space

How
Coordinates and terrain features are mapped into a latent representation using implicit neural networks (SIREN) to perform per-pixel conditioning of the weather forecasting process

Limitation
Hard to guarantee a real-world physical relationship between the topography and the target meteorological variable



Explicit Empirical Consistency Regulation



Motivation
Enforcing consistency over the mapping from altitude properties modeled as $f(\text{alt})$ to temperature properties modeled as $f(\text{temp})$, guides the model to align with empirical relational properties that directly originate from both topography and data, which are fastly learned due to the simplicity of the problem.

How
Spatial operator: $\phi_D : \mathcal{T}^{\text{in}}(\mathbb{R}_D) \rightarrow \mathcal{T}^{\text{target}}(\mathbb{R}_D)$

defining properties in the respective domain

Domain mapping: $\psi_{D \rightarrow G} : \mathcal{T}^{\alpha}(\mathbb{R}_D) \rightarrow \mathcal{T}^{\alpha}(\mathbb{R}_G)$

mapping a property in domain A to a property in domain B

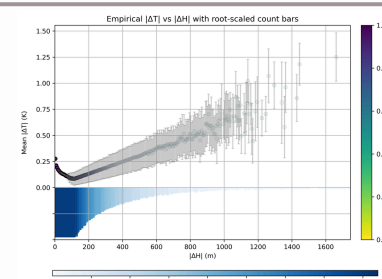
Consistency loss: $\mathcal{L}_{\text{ACL}}(p) = \mathcal{L}_{\text{metric}}(\psi_{H \rightarrow T}(\phi_H(H))_p, \phi_T(\tilde{T})_p)$

Instance #1 Histogram

Motivation
A simple relationship between temperature and altitude is the impact of the (absolute) differences

How
As a proof of concept the simple way of cuvefitting the histogram with $|dT|$ -Balls and $|dH|$ -Bins

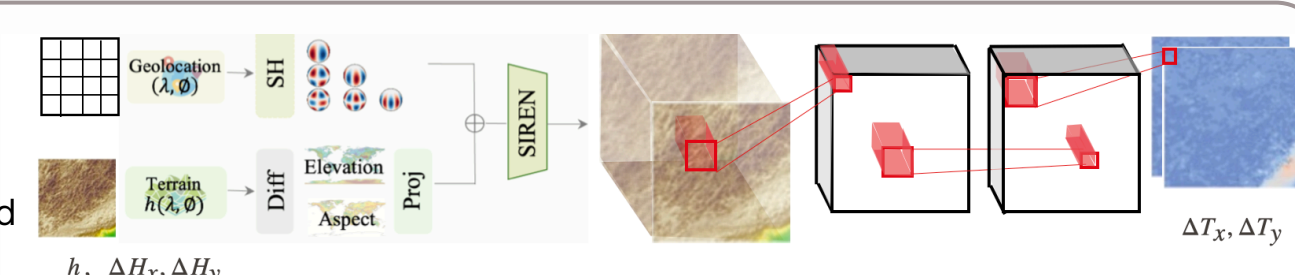
Limitation
Low variance in low-delta regions indicates high confidence with little need for regularization, while high variance in high-delta regions reduces effectiveness where it is most needed.



Instance #3 Space Dependent

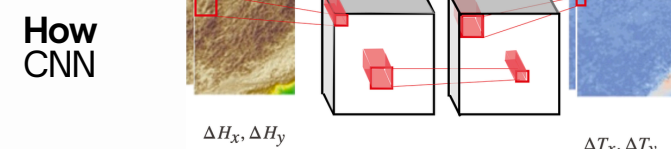
Motivation
Map from a geo-embedding to temperature difference

How
Prepend the GINR encoding and map to temperature difference



Instance #2 Space Agnostic

Motivation
Extend the idea of mapping from difference to difference, but condition on position rather than pseudo averaging as done for the histogram



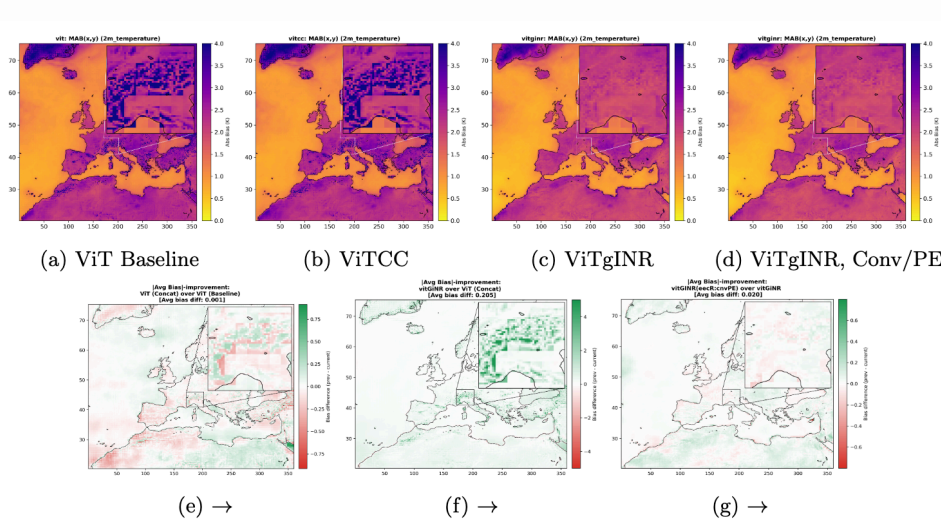
Limitation
Risk of learning signal being purely dominated by position rather than topography

Experiment Results

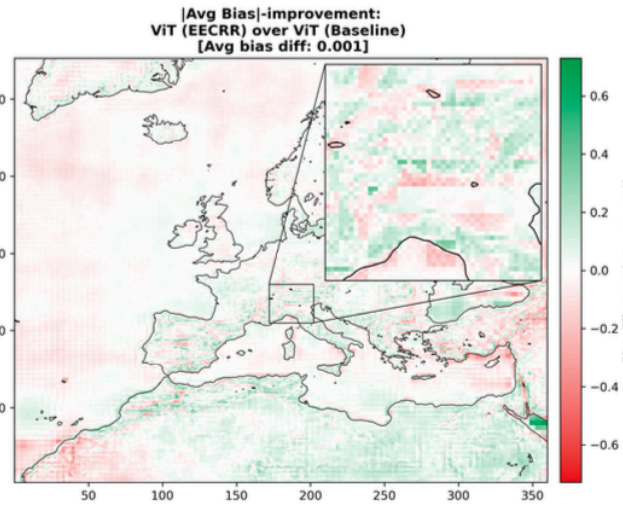
A Comparison of Different Methods. We compare the baseline with different ways of topography-informed learning, i.e., Simple fusion (ViTCC), Geo-INR (ViTGINR), EECR (HIST, Conv, Conv/PE)

Model	Train Loss	EECR($\lambda = 0.1$)	RMSE ↓	MAB
ViTGINR	MSE	HIST	2.4662	1.749
ViTGINR	MSE	Conv/PE	2.4718	1.756
ViTGINR	MSE	Conv	2.4795	1.761
ViTGINR	MSE	-	2.5028	1.776
ViT (Baseline)	MSE	HIST	2.7415	1.980
ViTCC	MSE	-	2.7507	1.980
ViT (Baseline)	MSE	-	2.7910	1.982

Remarkable Observations: Land-Sea Gap, Positional Encoding improvement, Greenland anomaly



Bias Comparison of Different Methods



ViT (EECR) vs ViT (Baseline)

Conclusion and Outlook

Summary

- Developed and evaluated methods to incorporate topography priors into neural forecasting models
- Identified performance trade-offs across encoding and fusion strategies
- Proposed EECR, achieving modest improvements by enforcing terrain-temperature consistency

Up next:

- Investigate advanced models: Fourier Neural Operators (FNOs), learnable Fourier features, and graph-based terrain embeddings
- Integrate additional data sources: land cover, vegetation indices, satellite-derived variables
- Apply multi-scale aggregation and attention mechanisms for richer spatial representation
- Further develop EECR into a general, scalable regularization framework for spatial learning

Related literature (Selection)

Xu, C., Sumbul, G., Mi, L., Zbinden, R., & Tuia, D. (2026). GeoFAR: Geography-informed frequency-aware super-resolution for climate data. In Proceedings of the International Conference on Learning Representations (ICLR 2026).

Nguyen, T., Jewik, J., Bansal, H., Sharma, P., & Grover, A. (2023). ClimateLearn: Benchmarking machine learning for weather and climate modeling. In Advances in Neural Information Processing Systems (NeurIPS 2023), Datasets and Benchmarks Track.

Lian, J., Huang, S., Shao, J., et al. (2024). TerraWind: A deep learning-based near-surface winds downscaling model for complex terrain regions. Geophysical Research Letters.

RuBwurm, M., Klemmer, K., Rolf, E., Zbinden, R., & Tuia, D. (2024). Geographic location encoding with spherical harmonics and sinusoidal representation networks. In Proceedings of the International Conference on Learning Representations (ICLR 2024).